



Geometric Calibration Verification
DMC01-0117



Geometric Calibration Verification Certificate

Digital Mapping Camera (DMC)

DMC Serial Number: **DMC01-0117**

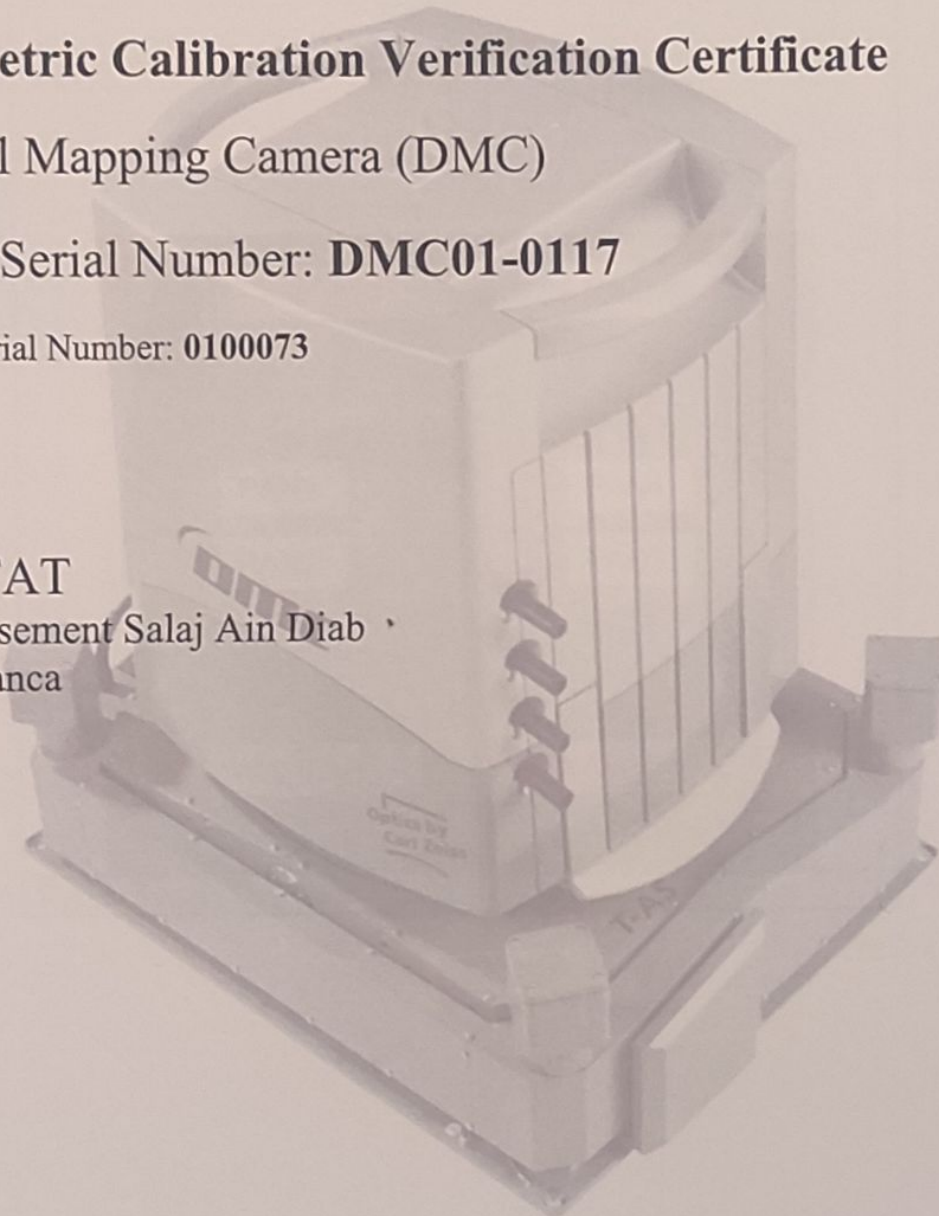
CBU Serial Number: **0100073**

For

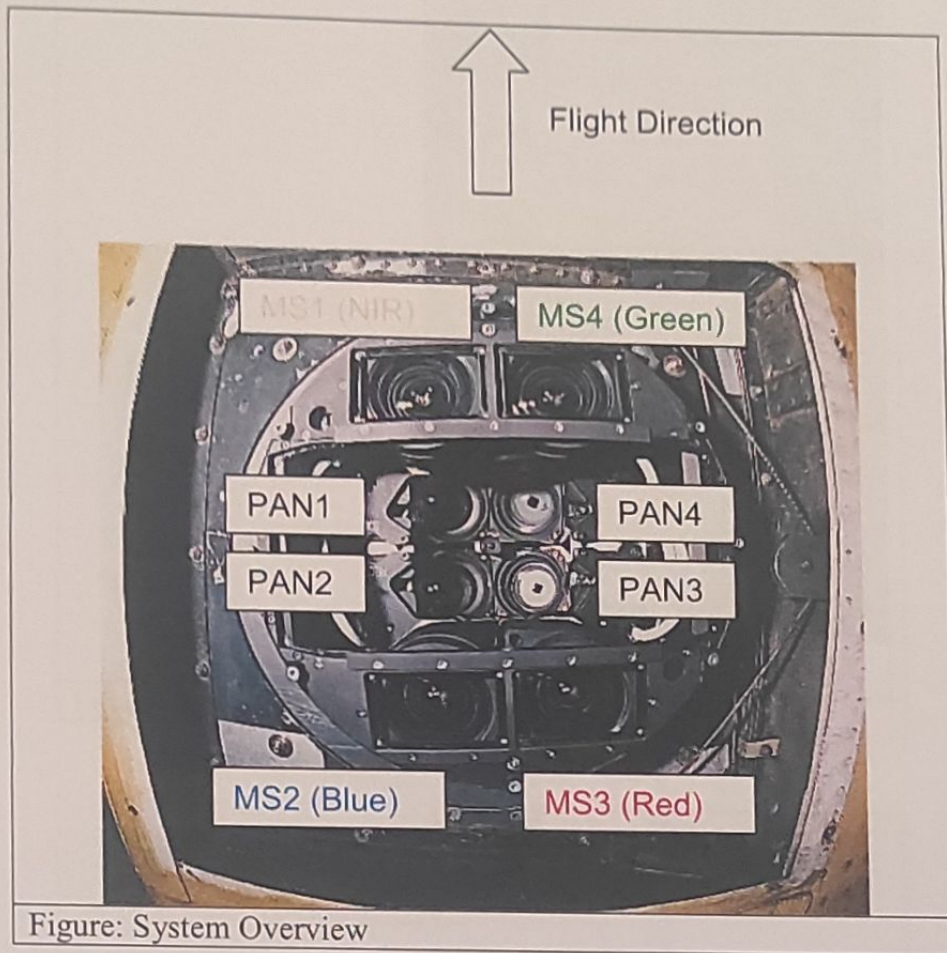
ETAFAT

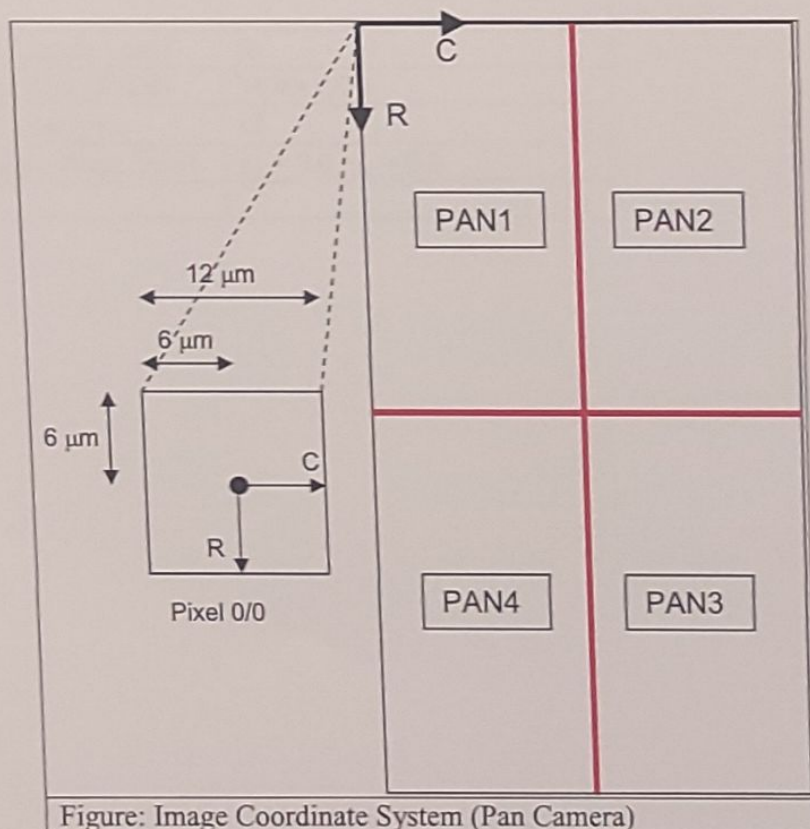
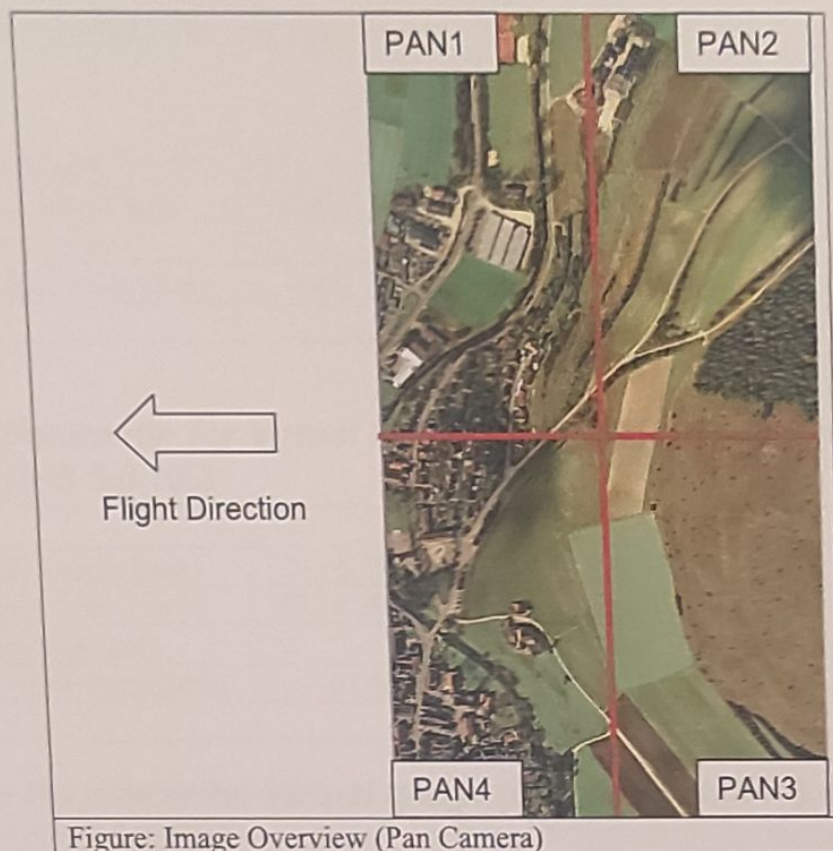
57 lotissement Salaj Ain Diab
Casablanca

Maroc



System Overview





Camera Parameter for Virtual Image (High Resolution)

Virtual Focal Length [mm]	120
Virtual Sensor Size [Pixel]	13824 x 7680
Virtual Pixel Size [μm]	12
Virtual Principle Point [mm]	$x_p = 0.0, y_p = 0.0$

Camera Parameter for Virtual Image (Color Resolution) before Version PPS 5.0.10.3

Virtual Focal Length [mm]	120 / 4.75
Virtual Sensor Size [Pixel]	3072 x 2048
Virtual Pixel Size [μm]	12
Virtual Principle Point [mm]	$x_p = -0.646, y_p = 0.646$

Camera Parameter for Virtual Image (Color Resolution) after Version PPS 5.1.10.3

Virtual Focal Length [mm]	30
Virtual Sensor Size [Pixel]	3456x1920
Virtual Pixel Size [μm]	12
Virtual Principle Point [mm]	$x_p = 0.0, y_p = 0.0$

Camera Serial Number and test flights

	Burn-in Flight: 11.03.2008	
	Calibration Test Flight: 24.10.2014	
Camera	Serial Number	Calib. Date
PAN1	00115569	27.02.2008
PAN2	00115573	27.02.2008
PAN3	00115590	27.02.2008
PAN4	00115593	29.02.2008
MS1 (NIR)	00114951	18.02.2008
MS2 (Blue)	00114957	20.02.2008
MS3 (Red)	00114972	18.02.2008
MS4 (Green)	00114975	15.02.2008

Camera Orientation PAN-Cameras (Test Flight 24.10.2014)

Camera (Serial Number)	X [m] (Accuracy)	Y [m] (Accuracy)	Z [m] (Accuracy)	Omega [Deg] (Accuracy)	Phi [Deg] (Accuracy)	Kappa [Deg] (Accuracy)
PAN1 (00115569)	0.064 (0)	-0.079 (0)	1000 (0)	17.954068 (0.001)	10.132992 (0.001)	86.675710 (0.001)
PAN2 (00115573)	-0.064 (0)	-0.079 (0)	1000 (0)	17.950471 (0.001)	-10.156391 (0.001)	92.942519 (0.001)
PAN3 (00115590)	-0.064 (0)	0.079 (0)	1000 (0)	-17.945762 (0.001)	-10.118467 (0.001)	-93.228623 (0.001)
PAN4 (00115593)	0.064 (0)	0.079 (0)	1000 (0)	-17.957047 (0.001)	10.141031 (0.001)	-86.738573 (0.001)

The data is connected to the virtual projection center of the virtual image.

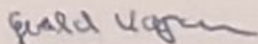
The above Platform calibration values are initial values and are liable to slight fluctuations between project images and between different projects. The rotation axes of the angles are (in this order)

Omega	x-Axis
Phi	y-Axis
Kappa	z-Axis

The results of the Platform calibration were generated with DMC Postprocessing SW (PPS), Version 6.7 and later, from Z/I Imaging sensor product suite.

Platform calibration verification performed by:

Z/I Employ Name:

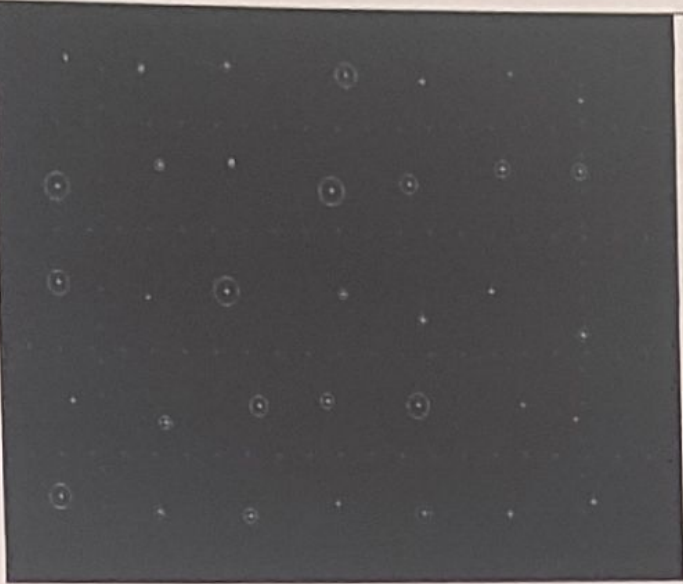


Dipl. Ing. Gerald Kapoun

18.11.2014

Date

Aerotriangulation Results (Test Flight 24.10.2014)

	Photo Scale	1:4210
	Flying Height [m]	500
	Flying Altitude [m]	715
	Run-Spacing [m]	483.8
	Base-Length [m]	153.6
	Number of Exposures	114
	Side-lap [%]	30
	End-lap [%]	60
	Terrain Height [m]	210
	Number of strips	6
	Photos in one strip	4 x 21 E-W 2 x 15 N-S
	Photos Used	114
	Control Points Used	5
	Check Points Used	30
	GSD [cm]	5



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Aerial triangulation statistic results:

Whole Block Sigma relative :		2.648 um		
Whole Block Sigma absolute :		2.700 um		
Parameter	X/...	Y/Phi	Z/K...	XY
RMS Control	0.029	0.012	0.021	0.022
RMS Check	0.028	0.024	0.037	0.026
RMS Limits	0.100	0.100	0.120	
Max Ground Residual	0.042	0.017	0.030	
Residual Limits	0.130	0.130	0.150	
Mean Std Dev Object				
RMS Photo Position	0.030	0.029	0.025	
RMS Photo Altitude	0.003	0.003	0.005	
Mean Std Dev Photo Po...				
Mean Std Dev Photo Atti...				
Key Statistics				
Sigma: 2.7 um				
RMS Image (x, y): 2.1, 2.0 um				
Number of iterations: 3				
Degrees of Freedom: 11486				
Gross Image Blunders: 0				
Gross Control Blunders: 0				
Image Blunders: 0				
Solution Status: Solution Successful				
Current Count		Cameras used: (1)		Project Settings
Control Points Used: 5		Camera Id	Len...	Grids
Check Points Used: 30		DMC	Off	Off
Photos Used: 114				
Photos Not Used: 0				
Image Points Used: 10594				
Linear: Meters Refraction: Off				
Angular: Degrees Curvature: Off				
Nouvelle Triangulation Francaise - Lambert				
Conformal Conic (1SP) (m)				

The results of the Aerotriangulation were generated with use of ImageStation Automatic Triangulation (ISAT), Version 2014 and later, from Intergraph photogrammetric product suite.

With this certificate we confirm that DMC01-0117 is within geometric accuracy.

Aerotriangulation performed by:

Z/I Employ Name:

Dipl. Ing. Gerald Kapoun

18.11.2014

Date